A STUDY ON THE DOUBLE EDGED WORLD OF ONLINE AFFAIRS AMONG STUDENTS OF THIRUVANANTHAPURAM DISTRICT

PROJECT REPORT 2026

Submitted to the University of Kerala in partial fulfilment of the requirements for the award of the Degree of Bachelor of Science in Statistics

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CERTIFICATE

This is to certify that the project under the title **"A STUDY ON DOUBLE EDGED WORLD OF ONLINE AFFAIRS AMONG STUDENTS OF THIRUVANANTHAPURAM DISTRICT "** is an authentic record of the work carried out by Ms.Asiya B, Ms.Shyna Shibu S, Ms.Fathima Hanna K, Ms.Anukrishna B S, Ms.Sivani S under my supervision and guidance, in partial fulfilment of the requirements for B.Sc. Degree in Statistics from the University of Kerala during 2023-2026.

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DECLARATION

We hereby declare that the project report titled **"A STUDY ON DOUBLE EDGED WORLD OF ONLINE AFFAIRS AMONG STUDENTS OF THIRUVANANTHAPURAM DISTRICT"** submitted to the University of Kerala in partial fulfilment of the requirements for the degree of Bachelor of Science in Statistics is a record of original work done by us under the supervision and guidance of Dr. Sheeja S S , Assistant Professor in Department of Statistics , Government College for Women and that no part of this project has been submitted earlier or elsewhere for any similar purpose.

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CONTENTS

CHAPTER 1: INTRODUCTION 1-3

1.1 : General introduction 1
1.2 : Aims and objectives of study 2
1.3 : Limitations of study 2-3
1.4 : Summary of study 3

CHAPTER 2: REVIEW OF LITERATURE 4-9

CHAPTER 3: METHODOLOGY 10-15

3.1 : Introduction 10
3.2 : collection and representation of data 10-11
3.3 : Mean and standard deviation 11
3.4 : ANOVA test for finding the significant relationship between means 11
3.4.1 :ANOVA test for finding the significant relationship between means of Gender And Academic performance Health care, E-commerce, Lack of social commitment and Daily routine 12
3.5 : Chi-Square test for independence of attributes 13
3.5.1 : Chi-Square test for independence for the relationship between the Gender and Online usage impact on Academic performance, Net banking ,Mental health,Transportation and Time consuming 13
3.5.2 : Chi-Square test for independence for the relationship between the mode of study and Awareness of Cyber fraud 14-15

CHAPTER 4: ANALYSIS USING DIAGRAMMATIC REPRESENTATION 16-26

4.1 : Introduction 16
4.2 : Age 16
4.3 : Area of residence 17
4.4 : Gender 17
4.5 : Mode of study 18
4.6 : Education and learning impact 19
4.7 :Daily routine and social commitment 20
4.8 : Online healthcare effectiveness 21

4.9 : Social media and entertainment impact 23
4.10 :Digital lifestyle (shopping, net banking and satisfaction) 25
4.11 : Conclusion 26

CHAPTER 5: ANALYSIS BASED ON SOME STATISTICAL METHODS 27-37
5.1 : Introduction 27
5.2 : Mean and standard deviation 27
5.3 : Chi square test for independence to find the relationship between gender and online usage impact on academic performance 28
5.4 : ANOVA test to find the significant relationship between gender and online usage Impact on health care 29
5.5 : ANOVA test to find the significant relationship between gender and online usage Impact on E-commerce 30
5.6 : Chi square test for independence to find the relationship between gender and online usage impact on net banking 31
5.7 : Chi square test for independence to find the relationship between gender and online usage impact on transportation 32
5.8 : Chi square test for independence to find the relationship between gender and online usage impact on mental health 32
5.9 : ANOVA test to find the significant relationship between gender and online usage Impact on lack of social commitment 33
5.10 : Chi square test for independence to find the relationship between gender and online usage impact on time consuming 34
5.11 : ANOVA test to find the significant relationship between gender and online usage Impact on daily routine 35
5.12 : Chi square test for independence to find the relationship between mode of study and awareness of cyber fraud 36

5.13: Suggestions 38
5.14: Conclusion 38

REFERENCES 39

APPENDIX 40-44

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Today, the online world has become a natural part of everyday life, especially among students. The growth of the internet, smartphones, and digital platforms has changed how students study, communicate, access services, and spend their time. What began as a helpful support system has now developed into a strong influence on the academic, personal, and social lives of students. While the online world offers many benefits, it also brings certain challenges, making it both helpful and harmful at the same time.

Students are among the most active users of online services because of their familiarity with technology and need for constant connection and update. Online platforms have made many daily activities easier and faster. Services such as net banking, online education, digital healthcare, transportation applications, and e-commerce have improved convenience and efficiency.

Net banking and digital payment systems allow students to manage money easily and independently. Online education has increased learning opportunities by providing access to virtual classes, study materials, and skill-based courses beyond the classroom. During the COVID-19 period, online learning played a major role in helping students continue their education.

The online world has also improved access to healthcare through online consultations and health related applications, enabling students to seek medical advice without much difficulty. Transportation applications have made travel more convenient and reliable, while e-commerce platforms allow students to purchase goods and services from their homes. These facilities have become a regular part of student life, saving time and effort.

At the same time, increased dependence on the online world has led to several concerns Spending long hours online, particularly on social media, can affect mental health by causing stress, anxiety, and emotional tiredness. Excessive screen time may encourage laziness and reduce physical activity, leading to unhealthy lifestyle habits. Online communication often replaces face-to-face interaction, which can weaken social relationships and reduce student's sense of responsibility towards family and society.

Another serious issue is the rise in online scams, cyber frauds, and digital safety risks. Lack of awareness and careless online behaviour can make students vulnerable to financial loss and misuse of personal information. Moreover, the online world is highly time-consuming, often resulting in distraction, poor time management, and reduced academic focus.

1.2 Aims and objectives of the study

The study aims to analyze the online activities and digital habits of students in Thiruvananthapuram. The following are the main objectives of the study:

- To analyze the demographic distribution of online usage patterns among respondents, focusing on daily hours spent and the nature of online activities.
- To evaluate the relationship between Gender and online usage impact, specifically determining if there is a significant difference in how online activities interfere with the daily routines of male and female students.
- To examine the level of awareness and trust regarding cyber fraud, assessing how scam calls and messages influence the reliability of online services across different modes of study (Offline and Online).
- To assess the perceived impact of digital engagement on academic performance and social commitments, identifying whether students believe online resources contribute positively or negatively to their productivity.
- To identify the barriers to the adoption of online utility services such as e-commerce, digital banking and telemedicine within the target population.

1.3 Limitations of the study

Some of the limitations of the study are

- Mostly the women students in colleges in Thiruvananthapuram are included in the study.
- The majority of the respondents are from 18-25 age groups.

- The study focuses only on selected aspects of online affairs so it may not cover all possible impacts of digital life on students.
- Also, it is difficult to determine how honestly the participants answered the items in the questionnaire ,thus may affect their answer choice

1.4 Summary of the Study

The present study consists of five chapters which includes the methods of data collection, various statistical tests and their inferences. The first chapter is an introductory chapter, where we include a general introduction about the topic "A Study on the double edged world of online affairs Among Students in Thiruvananthapuram". The general introduction discusses the significance of study and the factors of interest for selecting their topic. Then the first chapter includes the uses, awareness, aim and objective of the study. This chapter also contains some of the limitations that limited our study from actual scope of the study. The summary of the project is also included in the first chapter.

The second chapter includes "Review of literature", which presents a detailed overview of various previous studies related to the topic. It discusses the findings, methods and conclusions of different researchers on the impact of online activities among students. This chapter helps us to understand the positive and negative aspects identified in earlier studies and provides a base for the present study. It also highlights the research gap and justifies the need for conducting the current study.

The third chapter is fitted as "Methodology", which explains about the various methods used for the analysis of the study. It discusses the collection and representation of the data using various bar diagrams and pie charts. This chapter also includes explanation of mean and standard deviation of variables, one way ANOVA testing and testing for independence using Chi- Square.

In the fourth chapter, we have done the statistical analysis using the diagrams for some of the collected data. There we include an introduction to the chapter and then a total of charts and diagrams for the analysis of various factors.

In the fifth chapter we include "analysis based on some methods". There are three tests included in it. Two of them are based on Chi – Square distribution and one is ANOVA testing. At last references and appendix also attached.

CHAPTER 2

REVIEW OF LITERATURE

Waaswa Shafik (2026) studied Artificial Intelligence: Definitions and Perspectives. The study found that the AI is traced from early human-centred notions like "thinking machines" to modern, procedure based definitions. Shafik argues that intelligence can be described precisely enough to be simulated by machines, with the difference between natural and artificial systems lying only in their implementation biological versus mechanical. This view suggests that if cognition and behaviour are fully understood, they can be recreated in artificial devices. Ultimately, Shafik frames AI not as abstract philosophy but as a practical challenge of description and simulation, reshaping how we understand intelligence itself.

Adit Gupta (2026) studied student engagement with LinkedIn Learning in Indian higher education institutions. The research aimed to understand students' perceptions, attitudes, and the effectiveness of the platform in enhancing learning outcomes. The findings revealed that students generally had positive perceptions and were highly motivated to use the platform for skill development. Institutional support, proper curriculum integration, and guidance from educators were identified as key factors influencing its success. Additionally, features such as assessments, certifications, and personalized learning paths significantly improved student engagement and motivation.

Xuan Liu (2025) studied AI Awareness and Career Decision-Making Difficulties. The study shows how college students' awareness of AI affects their career choices. Liu finds that awareness can increase employment anxiety by highlighting job replacement risks, but it also motivates exploration and adaptability in the labour market. The research emphasizes that AI oriented career management reduces negative stress while strengthening positive exploration behaviours. Overall, Liu frames AI awareness as a double-edged sword that can transform stress into resilience when guided effectively.

Harsh Shukla (2025) examined the relationship between social media usage and academic engagement among students. The study found that social media platforms facilitate better communication, collaboration, and easy access to academic resources. Students were able to share knowledge, participate in discussions, and enhance their learning experiences through these platforms. However, excessive usage was associated with negative effects such as distraction, procrastination, and mental health concerns. The study emphasized the importance of maintaining a balanced and responsible use of social media for academic purposes.

Celia B.Fisher & Madeline Ford (2024) studied social media and LGBTQ+ Youth Development .The study examines how social media serves as both a supportive and harmful

space for LGBTQ+ adolescents. Through mixed methods with 406 participants, they identified themes of connection, refuge, and rejection, showing that online discrimination can intensify risks like depression and anxiety. At the same time, social media fosters belonging and exploration, making it a double-edged sword. Fisher and Ford stress the need for supportive strategies that balance these risks, helping strengthen resilience and protect youth mental health.

Rabindranath Karmakar and Biswajit Das (2024) analysed the impact of social networking sites such as Facebook and Instagram on student's academic performance. The study highlighted that these platforms serve as useful tools for communication, collaboration, and knowledge sharing among students. It also showed that students could benefit academically when using these platforms for educational purposes. However, excessive non-academic use resulted in reduced study time and lower academic performance. The researchers stressed the need for controlled and balanced usage to maximize benefits while minimizing negative impacts.

Muhammed Miftahul Amri (2023) studied ChatGPT in Medical Education and Research. The paper explores both the benefits and risks of large language models in medicine. Amri highlights how ChatGPT can aid students with information access, simulate patient interactions, and support researchers in drafting articles, but also warns of ethical concerns like cheating and misuse. Testing showed ChatGPT solved only one out of three Indonesian medical exam questions, revealing its current limitations, while many professionals struggled to distinguish AI-generated abstracts from real ones. The study concludes that ChatGPT holds promise for education and research but requires strong policies and ethical guidelines to ensure responsible use.

Mool Raj (2023) examined the relationship between social media usage, academic performance, and mental health among students. The findings indicated that social media plays an important role in enhancing communication and providing access to information and learning materials. Students benefited from networking and academic support through online platforms. However, overuse of social media was linked to increased stress, anxiety, and reduced academic focus. The study emphasized the importance of maintaining a healthy balance between online activities and academic responsibilities.

Frontiers in Psychology (2022) studied Leadership Task Transitions with Technology during COVID-19 in India. The study explores how digital tools shaped lives during the pandemic. Technology supported work, study, and connection, boosting resilience and motivation, but uneven access created barriers, especially for rural and lower-income groups. Common stresses included fear of infection, worry for loved ones, frustration, and disturbed sleep. The findings show technology as a double-edged sword, with self-efficacy mediating its impact on emotional well-being. The study concludes that reducing digital divides and strengthening psychological resources are key to building resilient communities post-COVID.

Kelum A. A. Gamage (2022) focused on Student engagement in online and hybrid learning environments, especially during the COVID-19 period. The study found that students were

able to adapt to digital modes of learning, although challenges in engagement and interaction were observed. Many students expressed a preference for hybrid learning, which combines both online and face-to-face teaching methods. The research highlighted that effective planning, interactive teaching strategies, and proper use of technology are essential to improve student engagement. It concluded that hybrid learning models can provide a more balanced and effective educational experience.

Felecia D'souza and Sita Shah (2021) studied social media and Medical Education. The article examines how social media platforms are transforming learning in the post-COVID era. Benefits include accessibility to experts, mentorship opportunities, global participation, and support networks, while also driving curriculum changes to address health inequities. However, risks involve superficial learning, professionalism concerns, psychological safety, and limited protections for content creators. These challenges highlight the double-edged nature of Some in medical education. The authors stress that the community must adapt quickly, critically evaluate innovations, and remain vigilant. By doing so, educators can maximize benefits while minimizing risks in this evolving digital landscape.

Upadhyaya and Vrinda (2021) studied Technostress among Indian University Students. The technostress is examined among Indian university students aged 18–28, highlighting a gap in research compared to the extensive focus on working professionals. With the growing reliance on digital tools in higher education such as learning management systems, MOOCs, and online assessments students are compelled to develop ICT skills, which can lead to stress. Using a sample of 673 students, the researchers validated the technostress instrument for academic contexts, finding moderate levels of stress overall. Importantly, the study revealed that technostress negatively impacts academic productivity, underscoring the need to address its effects in educational settings. This work contributes to the literature by extending technostress research to younger populations and emphasizing its relevance in academia. extending technostress research to younger populations and emphasizing its relevance in academia.

Hui luan and hollis lai (2020) studied challenges and future directions of big data and AI in education. Big data and AI are transforming education research, policy, and industry. Advances in educational technologies have made large-scale data collection and analysis possible, moving beyond experimental applications to widespread adoption. Key trends include assessment, individualized learning, and precision education, supported by model-driven analytics. However, results from educational analytics must be applied carefully to avoid misinterpretation. Policymakers should support lifelong learning, teacher training, and data protection, while industry and academia should collaborate closely. Finally, dialogue between technology and humanity is essential to ensure that AI and big data enhance pedagogy and learning in meaningful ways. (2020)

Shahida Parveen Kundan Singh and Pratibha Sagar (2020) conducted a study on Social media addiction among students in higher educational institutions in India. The study was carried out using a sample of 200 students and data were collected through a self-structured Social Media Scale. The findings revealed that excessive use of social networking sites such as WhatsApp, Instagram, and Facebook has both social and academic implications. While social media helps students stay connected and access information easily, overuse leads to

reduced social interaction, health issues, and poor academic performance. The study concluded that uncontrolled use of social media negatively affects students' learning outcomes and overall well-being.

Tras zeliha (2019) conducted study on internet addition and loneliness as predictors of internet gaming disorder in adolescents .This study looked at how internet gaming disorder (IGD) is related to internet addiction and loneliness among 406 teenagers in Ankara during the 2018– 2019 school year. The group included 305 boys and 101 girls, aged 14 to 18. Results showed that internet addiction strongly predicts IGD (ß = 0.54), while loneliness also predicts IGD but at a smaller level (ß = 0.11). Loneliness was also found to predict internet addiction (ß = 0.24, $p < 0.01$). In simple terms, teenagers who feel lonely are more likely to develop internet addiction, and those with higher internet addiction are more likely to show signs of gaming disorder.

Sharma R and Mehta A (2019) investigated the impact of digital learning platforms on academic achievement among undergraduate students in Delhi University. The study, conducted with a sample of 450 students across various disciplines, found that students who regularly engaged with e-learning platforms demonstrated significantly higher academic scores compared to their peers who relied solely on traditional classroom instruction. The researchers identified access to recorded lectures, interactive quizzes, and peer discussion forums as the primary factors contributing to improved outcomes. The study concluded that institutional investment in robust digital infrastructure and faculty training is critical for the successful integration of technology into the higher education curriculum.

S Rasheem and Alia Sha Alleman (2018) conducted study on mentor shape exploring the mentoring relationship of black women in doctoral programs. This phenomenological study, grounded in Black Feminist Thought, examined the lived mentoring experiences of 38 Black women in doctoral programs across U.S. universities. While prior research highlights mentorship and discrimination at the intersection of race and gender, little empirical work addresses how mentoring relationships form and what outcomes they produce for Black women in doctoral education. Findings revealed four central themes: organic evolvement, shared identity/mutual interest, personal transformation, and increased access and opportunity. These themes underscore the unique ways mentoring relationships support Black women's doctoral journeys. The study offers implications for future research, policy, practice, and education across disciplines.

Verma D K and Singh H (2018) analysed the influence of e-learning tools on the academic motivation of postgraduate students in Lucknow. Their research, involving 280 students enrolled in various postgraduate programmes, found that e-learning platforms contributed positively to intrinsic motivation by offering personalized learning experiences. Students appreciated the flexibility in accessing content at their own pace and the availability of diverse multimedia materials. The study also found that timely feedback mechanisms within these platforms played a critical role in sustaining motivation over longer academic periods. The authors recommended that universities regularly update their e-learning content and incorporate adaptive learning features to address individual learner differences.

Gregory D Moody And Benjamin Lowry (2017) conducted study on trust, distrust and ambivalence in online transaction .Trust and distrust are crucial in online relationships, yet their interplay is debated. Some view them as opposites, while others argue they are distinct constructs, leaving ambivalence largely unexplored. We propose a hybrid model where trust and distrust coexist as separate but related continua. Using polynomial regression and response surface analysis on data from 521 online consumers, we find strong support for this model. Results show that ambivalence has a small positive effect on consumers' intentions toward websites. Overall, trust and distrust are best understood as separate components with complex interrelations, opening new avenues for research.

Krishnan A and Menon S R (2017) examined The use of WhatsApp as an informal learning tool among undergraduate medical students in Bangalore. The study found that WhatsApp groups facilitated rapid sharing of academic materials, peer support, and timely clarification of doubts outside formal classroom hours. Students reported that the platform significantly reduced anxiety about examinations and improved their preparation by enabling constant access to study resources and group discussions. However, the research also identified challenges such as information overload, unverified content, and distraction as notable drawbacks. The study concluded that while WhatsApp holds potential as a supplementary learning tool, its use must be guided by institutional policies and faculty mentorship to ensure academic integrity and focused use.

Eriak E Smith (2016) conducted study on double edged sword undergraduate perceptions of social media in their learning .This study explores how undergraduate students perceive and use social media technologies (SMTs) in their learning through interviews and a large survey. It finds that students view social media as a "double-edged sword" that can both support and distract from learning. Many students intentionally decide how and when to use these tools for academic purposes. The results highlight that the effectiveness of social media depends on the learning context and its specific features. Student interactions are mainly focused on peer-to- peer and content engagement rather than communication with teachers. Overall, the study emphasizes the need to understand how social media can be meaningfully integrated into education.(2016)

Bhatia R and Kapoor S (2016 examined The influence of e-library resources on research productivity among postgraduate students at Delhi University. The study found that students with regular access to and training in the use of digital libraries, such as JSTOR, EBSCO, and PubMed, produced research papers of significantly higher quality and demonstrated greater information literacy. The research identified lack of awareness about available e-resources, inadequate institutional subscriptions, and insufficient information literacy training as the primary barriers preventing students from making full use of digital library facilities. The study recommended mandatory library orientation programmes at the commencement of postgraduate courses and increased investment in institutional journal subscriptions to support academic research.

Jayeon Lee(2015),conducted study on Effects of Journalists' Social Media on Audience Perception". Social media has become an important platform for news, and journalists use it to connect with people. This study looks at how journalists sharing personal things and interacting with users affects audience opinions. It found that these activities make journalists seem more friendly and relatable. However, too much interaction can make them look less professional. People's opinions about journalists also affect how they view the news they produce. So, social media can both help and harm a journalist's image.

Nambiar P and George T K (2015) examined The digital divide and its consequences for educational equity in rural Karnataka. Using a mixed methods approach combining household surveys and in depth interviews, the study found that students from households without internet access were at a significant educational disadvantage compared to their urban counterparts. The gap was most pronounced in access to reference materials, competitive exam preparation resources, and online scholarship information. Community-level digital literacy centers established under government programmes partially mitigated these disparities, though their impact was limited by irregular operating hours and insufficient trained personnel. The study called for coordinated multi-stakeholder efforts involving government, civil society, and the private sector to develop sustainable models for bridging the digital divide in rural educational contexts.

CHAPTER 3
METHODOLOGY

3.1 Introduction

In this chapter, we have described the methods and techniques used for the analysis of the data. Mainly three methods are used for the present study. First one is representation of data using graphical methods and arriving inferences based on that. The other two methods are ANOVA test for significance and the test for independence for various elements in the study. Each one of them is described briefly in the following sections.

3.2 Collection and representation of Data

Our study is conducted based on the "A Study on the Double Edged World Of Online Affairs Among Students Of Thiruvananthapuram district". So, our population is a finite and real population. Also, the data to be collected is the primary data. The data was collected using the questionnaire method. Before drafting the questionnaire, we asked our friends, which provided some direction in drafting the questionnaire. The questionnaire was then modified according to the result of pilot study. In order to collect the data, we select the sampling method, simple random sampling without replacement (SRSWOR). For that the size of the sample was determined. On an average there may be 5000 students of government colleges in Trivandrum district. So, we select 3% of the population, that is, 150 as the sample size. The data was collected from 20th January 2026 to 10th February 2026 via questionnaire and 150 students in Thiruvananthapuram district participated voluntarily.

After the collection of data, software excel is used for the data entry and analysis of the present study. The collected data were used for some kind of testing related to association. The other data were represented with the help of diagrams. For which different kinds of bar diagrams, pie diagrams and column diagrams were used. Different kinds of bar diagrams include simple bar diagrams, multiple bar diagrams and subdivided bar diagrams. In this study three types of bar diagrams were used.

Simple bar diagram, the simplest of the bar diagrams, used frequently in practice for the comparative study of two or more items or values of a single variable or in a single classification or category of data.

If we want to show an aggregate and also its divisions into various parts: a circle may be drawn, and divided into various sectors. This is called pie diagram. Pie diagrams enable us to make two types of comparison- Comparison of individual component and comparison of each component with total.

3.3 Mean And Standard Deviation

To examine how various factors such as Academic performance, Health care, E-commerce, Net banking, Transportation, Mental health, Lack of social commitment, Time consumption, and Daily routine affect respondents. Primary data were collected through a structured questionnaire. A total of 150 respondents participated in the survey, selected using a convenience sampling method. To analyze the data, statistical tools such as mean and standard deviation were used. The mean was calculated to determine the average response of participants for each variable using the formula :

$\bar{x} = \frac{\sum xi}{n}$, where x_i is the individual responses and n = 150

The standard deviation was computed to measure the variability of responses around the mean using the formula :

$$\sigma = \sqrt{\frac{\sum(xi - \bar{x})^2}{n-1}}$$

The results were presented in tabular form showing the mean and standard deviation for each variable to enable clear comparison and interpretation.

3.4 ANOVA test for finding the significant relationship between means

3.4.1 ANOVA test for finding the significant relationship between means of Gender and Health care, E-commerce, Lack of social commitment and Daily routine

The One-Way ANOVA technique was used in this study to examine whether there is a significant difference in the mean impact scores of online usage between male and female respondents with respect to Health care, E-commerce, Lack of social commitment and Daily routine. For each variable the total scores, sample sizes, means and standard deviations for both genders were calculated. To test this we take hypothesis as

$$H_0: \mu_{female} = \mu_{male}$$

$$H_1: \mu_{female} \neq \mu_{male}$$

Then the ANOVA equations used in analysis are :

1. Correction Factor, $CF = \frac{\Sigma(X)}{N}$

2. Total Sum of Squares, $SST = \Sigma X^2 - CF$ with degrees of freedom = k-1 where , k is the number of levels

3. Sum of Squares Between Groups, $SSB = \Sigma \left(\frac{Xi^2}{ni}\right) - CF$ with degrees of freedom = N-k

   where X_i = Total of each group and n_i = Number of observations in each group

4. Sum of Squares of Error, SSE = SST- SSB with degrees of freedom = N-1

5. Mean Sum of Squares:

   Mean Sum of Square Between Groups, $MSB = \frac{SSB}{k-1}$

   Mean Sum of Errors, $MSE = \frac{SSE}{N-k}$

6. F-Statistic, $F = \frac{MSB}{MSE}$

ANOVA Table

Source of variation	Sum of Squares	Degrees of freedom	Mean sum of squares	F-statistic
Between groups	SSB	k-1	MSB	F- value
Error	SSE	N-k	MSE	
Total	SST	N-1		

If $F > F_\alpha$ we reject Ho

If $F < F_\alpha$ we accept Ho

3.5 Chi-Square test for Independence of Attributes

3.5.1 Chi-Square test for independence for the relationship between the Gender and Online usage impact on Academic performance, Net banking, Transportation, Mental health and Time consuming.

The Chi-Square test for Independence was used to determine whether there is a significant association between two categorical variables such as Gender and Online usage impact on various factors. For each analysis, a contingency table of observed frequencies was prepared. The table is given as below:

Gender	No impact	Slight impact	Moderate impact	Strong impact	High impact	Total
Female	O(E)	O(E)	O(E)	O(E)	O(E)	
Male	O(E)	O(E)	O(E)	O(E)	O(E)	
Total						

where , O =Observed frequency

E= Expected frequency

We have to test :

Ho: Categorical variables are independent

H1: Categorical values are dependent

The expected frequency, $E = \frac{(\text{Row Total x Column Total})}{Grand\ total}$

The $\chi^2 = \sum(\frac{(O-E)^2}{E})$ which is distributed as a Chi-Square variate with (r-1) degrees of freedom

We reject Ho if $\chi^2 > \chi^2_\alpha$

We accept Ho if $\chi^2 < \chi^2_\alpha$

3.5.2 Chi-Square test for independence for the relationship between the Mode of study and Awareness of Cyber fraud.

The Chi-Square test for Independence was used to determine whether there is a significant association between two categorical variable Mode of study and Awareness of Cyber fraud. For each analysis, a contingency table of observed frequencies was prepared. The table is given as below:

Mode of study	Highly aware	Not aware	Slightly aware	Total
Hybrid	O(E)	O(E)	O(E)	
Offline	O(E)	O(E)	O(E)	
Online	O(E)	O(E)	O(E)	
Total				

where , O =Observed frequency

E= Expected frequency

We have to test :

Ho: Categorical variables are independent

H1: Categorical values are dependent

The expected frequency, $E = \frac{(\text{Row Total x Column Total})}{Grand\ total}$

The $\chi^2 = \sum \frac{(O-E)^2}{E}$ which is distributed as a χ^2 - variate with (r-1) degrees of freedom

We reject Ho if $\chi^2 > \chi^2_\alpha$

We accept Ho if $\chi^2 < \chi^2_\alpha$

CHAPTER 4

ANALYSIS USING DIAGRAMMATIC REPRESENTATION

4.1 Introduction

In this chapter, we first classify some of the data regarding general aspects of online affairs among students of colleges of Thiruvananthapuram district. Then these classified data are represented using bar diagrams and pie charts for easy interpretation. We mainly included some of the data related to the positive and negative side of online in this chapter. Finally, we pointed out the inference obtained from this analysis.

4.2 Age

The frequency distribution of showing age of respondent participated in the study is given below:

Age	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 1

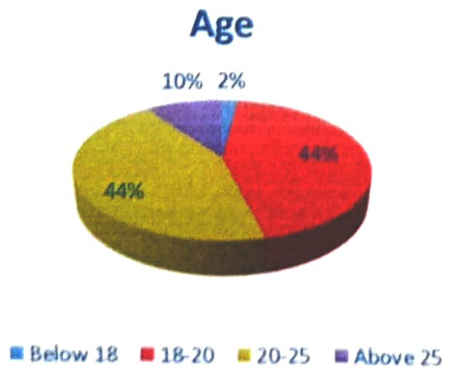


Fig 1

4.3 Area of residence

The frequency distribution of area of residence of respondents participated in the study is given below:

Area of residence	Frequency	Percentage
Rural area	79	52.7
Urban area	71	47.3
Total	150	100

Table 2

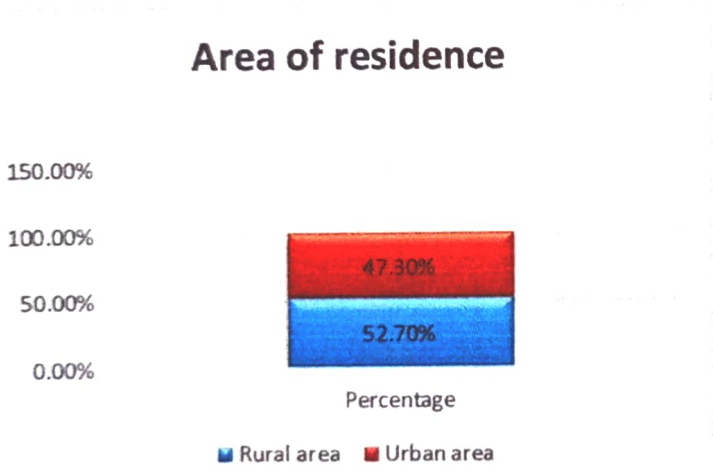


Fig 2

4.4 Gender

The frequency distribution of gender of respondents participated in the study are given below:

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 3

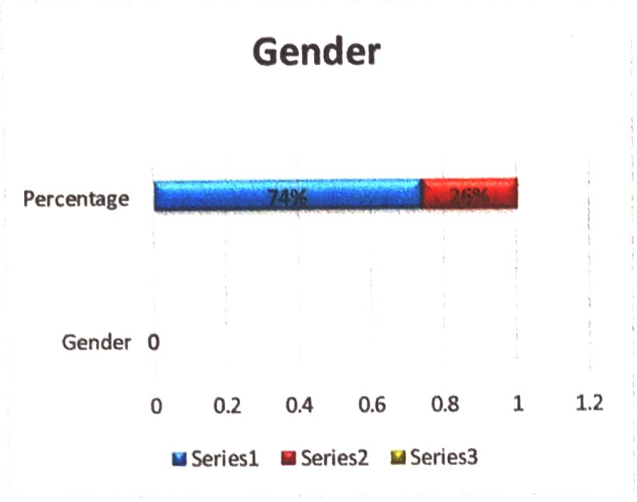


Fig 3

4.5 Mode of study

The frequency distribution of mode of study of the respondent participated in the study are given below:

Mode of study	Frequency	Percentage
Offline	100	66.7
Hybrid	31	20.7
Online	19	12.7
Total	150	100

Table 4

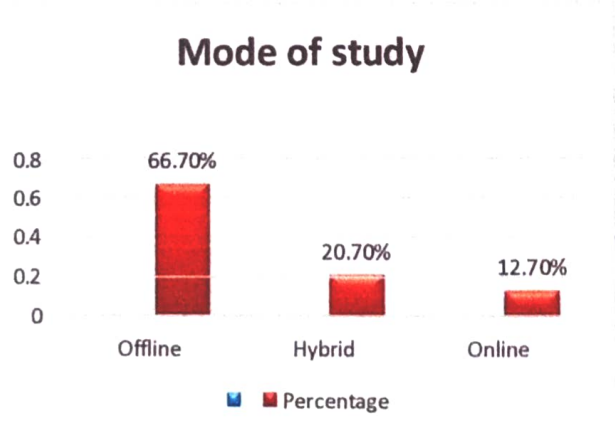


Fig 4

4.6 Education and Learning Impact

The frequency distribution of showing impact of online affairs in education among students is given below:

Gender wise

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 5

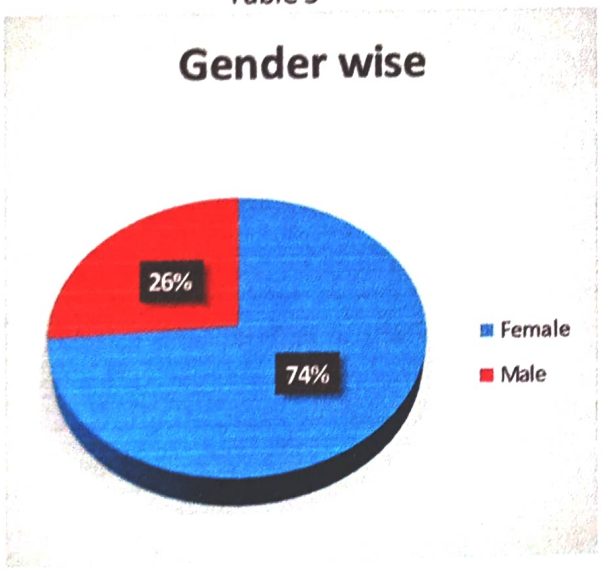


Fig 5

The data shows a clear female majority, with 74% females compared to 26% males, indicating a strong gender imbalance toward females.

Age wise

Age group	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 6

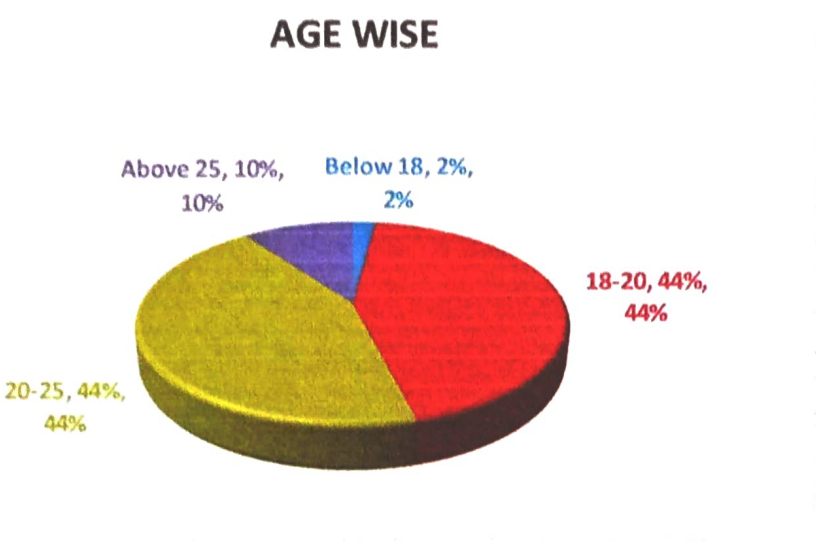


Fig 6

The data shows that most respondents fall within the 18–20 (44%) and 20–25 (44%) age groups, with a smaller proportion above 25 (10%) and very few below 18 (2%).

4.7 Daily Routine and Social Commitment

The frequency distribution of showing impact of online affairs in daily schedule and social involvement among students is given below:

Gender Wise

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 7

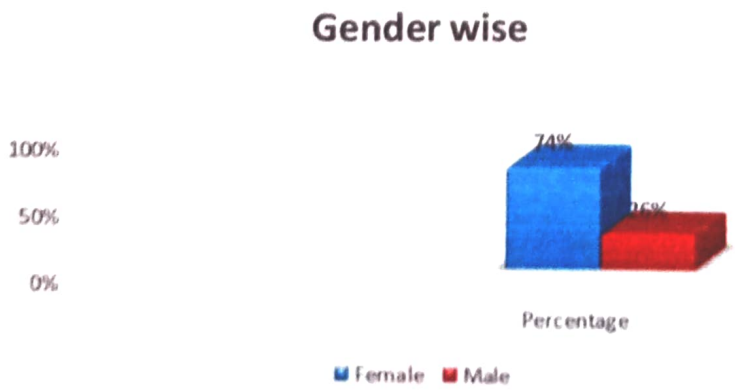


Fig 7

The data indicates that females constitute the majority at 74% (111 out of 150), while males make up 26% (39 out of 150), showing a clear gender imbalance toward females.

Age wise

Age group	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 8

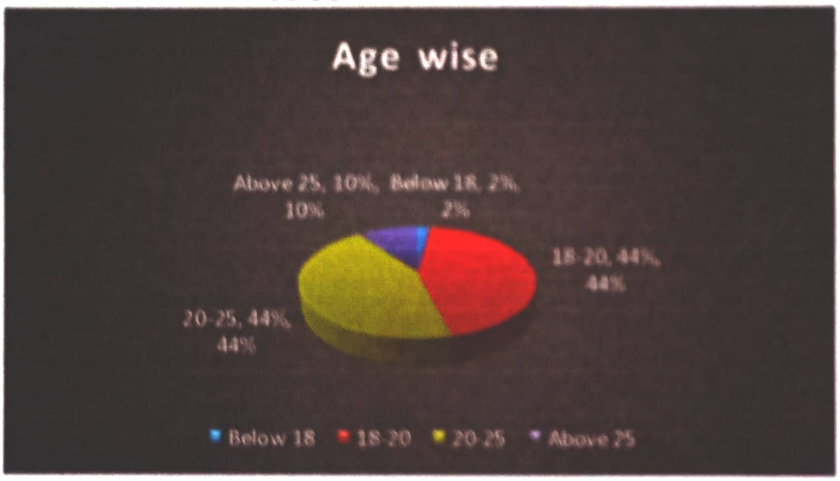


Fig 8

The data indicates that most respondents are in the 18–20 (44%) and 20–25 (44%) age groups, while fewer are above 25 (10%) and very few are below 18 (2%)

4.8 Online Healthcare Effectiveness

The frequency distribution of showing impact of online affairs in healthcare among students is given below:

Gender wise

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 9

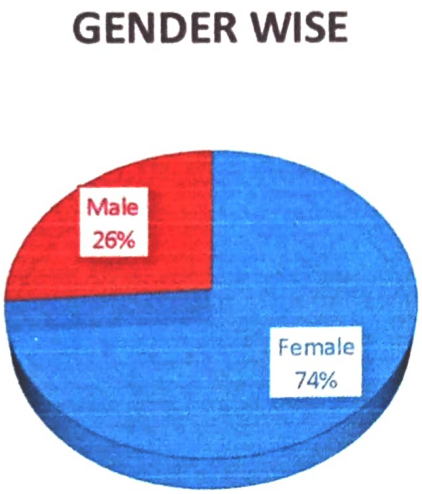


Fig 9

The distribution shows that females account for 74% (111) of the total while males represent 26% (39), indicating a strong female majority in the group.

Age wise

Age group	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 10

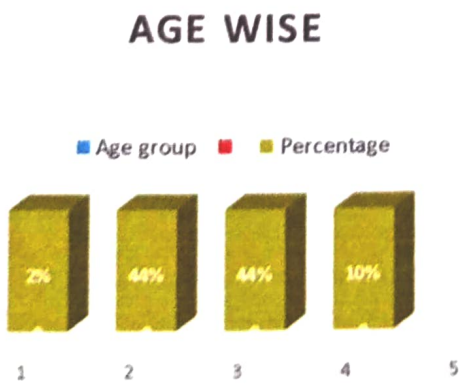


Fig 10

The table shows that most respondents are aged 18–25 (88%), with very few below 18 (2%) and 10% above 25.

4.9 Social media and Entertainment impact

The frequency distribution of showing influence of social media among students is given below:

Gender wise

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 11

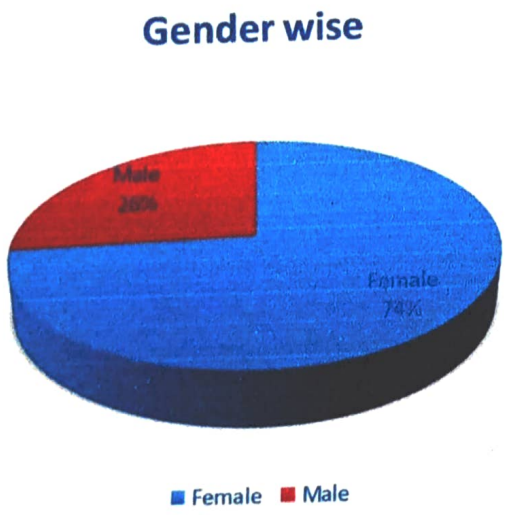


Fig 11

The table shows that females (74%) form the majority ofrespondents, while males account for 26% of the total sample.

Age wise

Age group	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 12

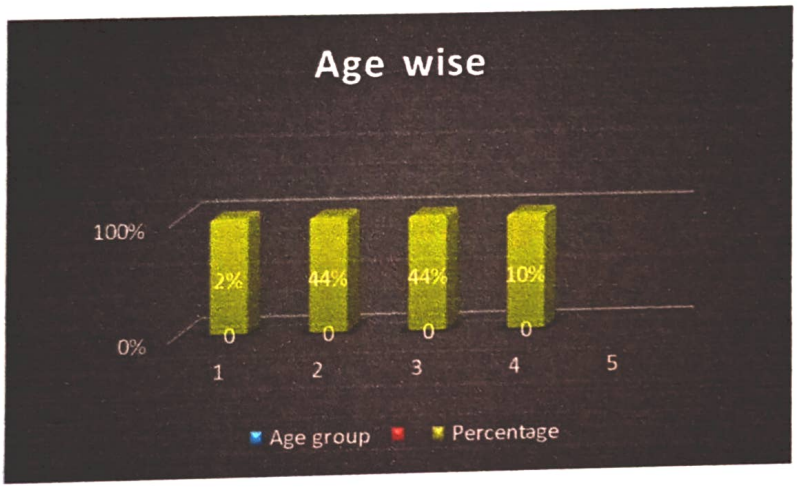


Fig 12

The data shows that respondents are predominantly aged 18–25 (88%), with equal representation in the 18–20 and 20–25 groups (44% each), indicating a strong concentration of young adults in the sample.

4.10 Digital lifestyle (shopping, net banking and satisfaction)

The frequency distribution of showing impact of online affairs in lifestyle among students is given below:

Gender wise

Gender	Frequency	Percentage
Female	111	74
Male	39	26
Total	150	100

Table 13

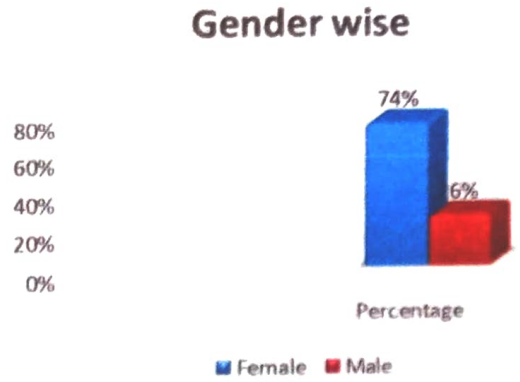


Fig 13

The data shows that the majority of respondents are female (74%), while males constitute only 26% of the total sample.

Age wise

Age group	Frequency	Percentage
Below 18	3	2
18-20	66	44
20-25	66	44
Above 25	15	10
Total	150	100

Table 14

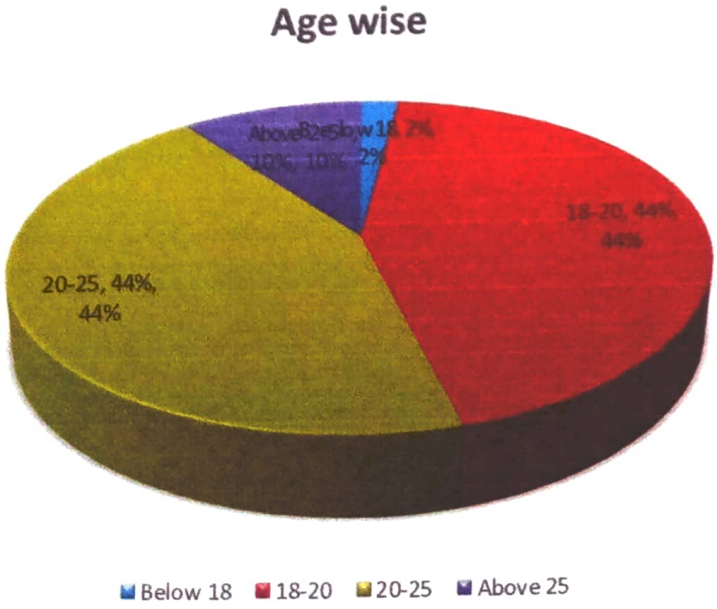


Fig 14

The data indicates that most respondents belong to the 18–25 age group (88%), with equal proportions in 18–20 and 20–25 (44% each), showing a strong concentration of young adults.

4.11 Conclusion

- The majority of respondents (88%) belong to the 18–25 age group, showing that the study mainly represents young adults.
- There is a significant gender imbalance, with females forming 74% of the sample and males only 26%.
- Respondents are almost equally distributed between rural (52.7%) and urban (47.3%) areas, indicating balanced geographical representation.
- Most students prefer the offline mode of study (66.7%), while fewer use hybrid (20.7%) and online (12.7%) modes.
- Online affairs have a noticeable impact on students' education and learning, affecting their academic activities.
- Students' daily routines and social commitments are influenced by online engagement, indicating changes in time management and interaction patterns.
- Online platforms play a role in healthcare awareness and accessibility, though the extent varies among students.
- Social media and entertainment significantly influence students, shaping their behaviour, communication, and leisure activities.
- The use of online services like shopping and net banking is common, reflecting a growing digital lifestyle among students.

CHAPTER 5

ANALYSIS BASED ON SOME STATISTICAL METHODS

5.1 Introduction

For the study on double edged world of online affairs among students of Thiruvananthapuram, we use some statistical test based on the ANOVA and chi square distributions. First section reveals the significant relationship between gender and factors also and in the second section we tested the independence between gender and factors. And the conclusion obtained from these statistical tests are also listed.

5.2 MEAN AND STANDARD DEVIATION

The mean was calculated to determine the average response of participants for each variable using the formula:

$$\bar{x} = \frac{\sum_{i=1}^{n} xi}{n}$$, where x_i is the individual responses and n=150

The standard deviation was computed to measure the variability of responses around the mean using the formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(xi-\bar{x})^2}{n-1}}$$

The results were presented in tabular form showing the mean and standard deviation for each variable to enable clear comparison and interpretations

	Mean	Standard Deviation
Academic performance	3.33	1.24
Health care	2.71	1.07
E-commerce	2.97	1.20
Net banking	3.35	1.33
Transportation	3.03	1.26
Mental health	2.70	1.22
Lack of social commitment	2.85	1.18
Time consuming	3.39	1.26
Daily routine	3.06	1.21

5.3 Chi square test for independence to find the relationship between gender and online usage impact on Academic performance

We have to test

H_0 : gender and the online usage impact on academic performance are independent

H_1 : gender and the online usage impact on academic performance are dependent

The contingency table is given as below

Gender	No impact	Slight impact	Moderate	Impact	High impact	Total
Female	11(9.62)	25(19.98)	32(28.12)	28(30.34)	15(22.94)	111
Male	2(3.38)	2(7.02)	6(9.88)	13(10.66)	16(8.06)	39
Total	13	27	38	41	31	150

Chi square statistic is ,

$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$

$=18.9357$

From the tables of chi square distribution $\alpha=0.05$ and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 > 9.488$, we reject H_0

Hence at 5% level of significance it may conclude that gender and the online usage impact on academic performance are dependent.

5.4 ANOVA test to find the significant relationship between Gender and Online Usage Impact on Health Care

We have to test

H_0:There is no significant difference in the mean of online usage impact on health care and gender

H_1:There is significant difference in the mean of online usage impact on health care and gender

5.3 Chi square test for independence to find the relationship between gender and online usage impact on Academic performance

We have to test

H_0 : gender and the online usage impact on academic performance are independent

H_1 : gender and the online usage impact on academic performance are dependent

The contingency table is given as below

Gender	No impact	Slight impact	Moderate	Impact	High impact	Total
Female	11(9.62)	25(19.98)	32(28.12)	28(30.34)	15(22.94)	111
Male	2(3.38)	2(7.02)	6(9.88)	13(10.66)	16(8.06)	39
Total	13	27	38	41	31	150

Chi square statistic is ,

$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$

$=18.9357$

From the tables of chi square distribution $\alpha=0.05$ and for 4 degrees of freedom we have tabled value 9.488

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5.4 ANOVA test to find the significant relationship between Gender and Online Usage Impact on Health Care

We have to test

H_0:There is no significant difference in the mean of online usage impact on health care and gender

H_1:There is significant difference in the mean of online usage impact on health care and gender

Gender	Sum of scores	Sample size	mean	Standard deviation
Female	292	111	2.6306	1.0351
Male	114	39	2.9231	1.1329
total	406	150	2.7067	1.0653

Number of levels, k=2

The correction factor, CF $=\frac{T2}{N}=\frac{406^2}{150}=$ 1098.9067

Total sum of squares ,SST = $\sum x^2$-CF = 1268-1098.9067 =169.0933

Where x^2 is 150 ×$((2.70)^2+(1.06)^2)$

Sum of squares between groups, SSB = $\left[\frac{(\sum x1)2}{n1}+\frac{(\sum x2)2}{n2}\right]-$ CF

$$=\left[\frac{292^2}{111}+\frac{114^2}{39}\right]-1098.91=2.47$$

Sum of squares of errors, SSE = SST-SSB = 169.09-2.47=166.63

Mean sum of squares between groups , MSB = SSB/df =2.47/1=2.47

Mean sum of squares of error, MSE = SSE/df=166.63/148=1.13

Now the F-statistic is , F=MSB/MSE = 2.47/1.13 = 2.19

ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F statistic
Between groups	2.47	k-1=1	2.47	2.19
Error	66.63	n-k=148	1.13	
total	169.09	n-1=149		

F_α=3.91,where α=0.05 with df (1,148)

F< F_α,so we accept H0

Hence there is no significant difference in the mean of gender and online usage impact on health care

5.5 ANOVA test to find the significant relationship between Gender and Online Usage Impact on E- commerce

We have to test

H_0:There is no significant difference in the mean of online usage impact on E-commerce and gender

H_1:There is significant difference in the mean of online usage impact on E- commerce and gender

Gender	Sum of scores	Sample size	mean	Standard deviation
Female	320	111	2.88	1.19
Male	125	39	3.21	1.22
total	445	150	2.97	1.20

Number of levels, k=2

The correction factor, CF $=\frac{T2}{N}=\frac{4452}{150}=\frac{198,025}{150}$ =1320.17

Total sum of squares ,SST = $\sum x2$-CF = 1535-1320.17 =214.83

Where x^2 is 150 ×$((2.97)^2)$+$((1.20)^2)$

Sum of squares between groups, SSB = $\left[\frac{(\sum x1)2}{n1}+\frac{(\sum x2)2}{n2}\right]-CF$

$$=\left[\frac{320^2}{111}+\frac{125^2}{39}\right]-1320.17=[922.52-400.64]-1320.17=3$$

Sum of squares of errors, SSE = SST-SSB = 214.83 – 3 = 211.84

Mean sum of squares between groups , MSB = SSB/df =3/1=3

Mean sum of squares of error, MSE = SSE/df = 211.84 /148 = 1.43

Now the F-statistic is , F= MSB/MSE = 3/1.43 = 2.09

ANOVA Table

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F statistic
Between groups	3	k-1=1	3	2.09
Error	211.84	n-k=148	1.43	
total	214.84	n-1=149		

F_α=3.91,where α=0.05 with df (1,148)

F< F_α,so we accept H_0

Hence there is no significant difference in the mean of gender and online usage impact on E-commerce

5.6 Chi square test for independence to find the relationship between gender and online usage impact on net banking

We have to test

H_0 : gender and the online usage impact on net banking are independent

H_1 : gender and the online usage impact on net banking are dependent

The contingency table is given as below

Gender	No impact	Slight impact	Moderate	Impact	High impact	Total
Female	17(14.06)	17(14.80)	31(28.12)	25(26.64)	31(27.38)	111
Male	2(4.94)	3(5.20)	7(9.88)	11(9.36)	16(9.62)	39
Total	19	20	38	36	37	150

Chi square statistic is ,

$$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$$

=10.863

From the tables of chi square distribution α=0.05 and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 > 9.488$, we reject H_0

Hence at 5% level of significance it may conclude that gender and the online usage impact on net banking are dependent.

5.7 Chi square test for independence to find the relationship between gender and online usage impact on Transportation

We have to test

H_0 : gender and the online usage impact on transportation are independent

H_1 : gender and the online usage impact on transportation are dependent

The contingency table is given as below

Gender	No impact	Slight impact	Moderate	Impact	High impact	Total
Female	19(16028)	21(19.98)	38(36.26)	21(21.46)	12(17.02)	111
Male	3(5.72)	6(7.02)	11(12.74)	8(7.54)	11(5.98)	39
Total	22	27	49	29	23	150

Chi square statistic is ,

$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$

=8.002

From the tables of chi square distribution α=0.05 and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 < 9.488$, we accept H_0

Hence at 5% level of significance it may conclude that gender and the online usage impact on transportation are independent

5.8 Chi-square test for independence to find the relationship between Gender and Online usage impact on Mental health.

We have to test,

H_0 : Gender and the online usage impact on mental health are independent.

H_1 : Gender and the online usage impact on mental health are dependent.

The contingency table is given as below,

Gender	No impact	Slight impact	Moderate	Strong impact	High impact	Total
Female	25(24.42)	22(22.20)	33(35.52)	22(19.98)	9(8.88)	111
Male	8(8.58)	8(7.80)	15(12.48)	5(7.02)	3(3.12)	39
Total	33	30	48	27	12	150

Chi square statistic is,

$$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$$

$$= 1.539$$

From the tables of chi square distribution $\alpha=0.05$ and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 < 9.488$, we accept H_0

Hence at 5% level of significance it may conclude that gender and the online usage impact on mental health are independent.

5.9 ANOVA test to find the significant relationship between Gender and Online usage impact on Lack of social commitment .

We have to test,

H_0 : There is no significant difference in the mean of online usage impact on lack of social commitment and gender.

H_1 : There is significant difference in the mean of online usage impact on lack of social commitment and gender.

Gender	Sum of scores	Sample size	Mean	Standard deviation
Female	304	111	2.74	1.17
Male	123	39	3.15	1.16
Total	427	150	2.85	1.18

Number of levels, k =2 ,

The correction factor, $CF = \frac{T^2}{N} = \frac{427^2}{150} = 1215.53$

Total sum of squares, SST= $\sum x^2$ - CF = 1423-1215.53 =207.47

where X^2 is 150 x $((2.85)^2+(1.18)^2)$ = 1423

Sum of squares between groups, SSB = $\left[\frac{(\sum X1)^2}{n1}+\frac{\sum(X2)^2}{n2}\right]$ - CF

$$=\left[\frac{304^2}{111}+\frac{123^2}{39}\right]-1215.53$$

$$= 4.97$$

Sum of squares of errors , SSE = SST – SSB = 207.47 – 4.97 = 202.50

Mean sum of squares between groups , MSB = SSB /df = 4.97/1 = 4.97

Mean sum of squares of error , MSE = SSE/df = 202.50/148 = 1.37

Now the F -statistic is , F = MSB/MSE = 4.97/1.37 = 3.63

ANOVA Table

Source of variation	Sum of Squares	Degrees of freedom	Mean sum of squares	F – statistic
Between groups	4.97	k-1=1	4.97	3.635
Error	202.50	N-k=148	1.37	
Total	207.47	N-1 =149		

F_α= 3.91 ,Where α= 0.05 with df (1,148)

F <F_α so we accept H_0

Hence there is no significant difference in the mean of gender and online usage impact on lack of social commitment.

5.10 Chi-square test for independence to find the relationship between Gender and Online usage impact on Time consuming.

We have to test ,

H_0 :Gender and the online usage impact on time consuming are independent.

H_1:Gender and the online usage impact on time consuming are dependent.

The contingency table is given as below,

Gender	No impact	Slight impact	Moderate	Strong impact	High impact	Total
Female	10(9.62)	22(18.50)	28(28.86)	27(27.38)	24(26.64)	111
Male	3(3.38)	3(6.50)	11(10.14)	10(9.62)	12(9.36)	39
Total	13	25	39	37	36	150

Chi square statistic is,

$$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$$

$$= 3.730$$

From the tables of chi square distribution $\alpha=0.05$ and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 < 9.488$, we accept H_0

Hence at 5% level of significance it may conclude that gender and the online usage impact on time consuming are independent.

5.11 ANOVA test to find the significant relationship between Gender and Online usage impact on Daily routine.

We have to test,

H_0 : There is no significant difference in the mean of online usage impact on daily routine and gender.

H_1 : There is significant difference in the mean of online usage impact on daily routine and gender.

Gender	Sum of scores	Sample size	Mean	Standard deviation
Female	330	111	2.97	1.16
Male	129	39	3.31	1.32
Total	459	150	3.06	1.21

Number of levels,k =2 ,

The correction factor, CF = $\frac{T^2}{N} = \frac{4592}{150} = 1404.54$

Total sum of squares, SST= $\sum X^2$ - CF = 1623-1404.54 = 218.46

where X^2 is 150 x $((3.06)^2+(1.21)^2)$=1623

Sum of squares between groups, SSB = $\left[\frac{(\sum X1)^2}{n1}+\frac{\sum(X2)^2}{n2}\right]$ - CF = $\left[\frac{330^2}{111}+\frac{129^2}{39}\right]-1404.54$

= 3.23

Sum of squares of errors, SSE = SST – SSB = 218.46 – 3.23 = 215.23

Mean sum of squares between groups, MSB = SSB / df = 3.23/1 = 3.23

Mean sum of squares of error, MSE = SSE/df = 215.23/148 = 1.45

Now the F -statistic is, F = MSB/MSE = 3.23/1.45 = 2.22

ANOVA Table

Source of variation	Sum of Squares	Degrees of freedom	Mean sum of squares	F – statistic
Between groups	3.23	k-1=1	3.23	2.22
Error	215.23	N-k=148	1.45	
Total	218.46	N-1 =149		

F_α= 3.91 ,where α= 0.05 with df (1,148)

$F < F_\alpha$ so we accept H_0

Hence there is no significant difference in the mean of gender and online usage impact on daily routine.

5.12 Chi-square test for independence to find the relationship between Mode of study and Awareness of cyber fraud.

We have to test ,

H_0 : Mode of study and awareness of cyber fraud are independent.

H_1: Mode of study and awareness of cyber fraud are dependent.

The contingency table is given as below,

Mode of study	Highly aware	Not aware	Slightly aware	Total
Hybrid	10(7.65)	1(2.69)	20(20.67)	31
Offline	20(24.67)	10(8.67)	70(66.67)	100
Online	7(4.69)	2(1.65)	10(12.67)	19
Total	37	13	100	150

Chi square statistic is,

$$\chi^2 = \sum \frac{(O-E)^2}{E} \sim \chi^2_{\alpha(r-1)}$$

$$= 4.838$$

From the tables of chi square distribution $\alpha=0.05$ and for 4 degrees of freedom we have tabled value 9.488

Hence $\chi^2 < 9.488$, we accept H_0

Hence at 5% level of significance it may conclude that mode of study and awareness of cyber fraud are independent.

5.13 Suggestions

Our study is useful for future projects and helps in understanding the digital habits of students in Thiruvananthapuram. It identifies that certain online services, like net banking, have a dependent relationship with gender, suggesting that targeted digital literacy programs would be helpful. Since the mode of study did not change the level of cyber fraud awareness, it is essential to develop a universal security curriculum for all students. Educational institutions can use these findings to encourage healthy online habits and balanced digital wellness among the youth. Finally, future studies could expand this research to different districts to see if these online behaviours remain consistent across the state.

5.14 Conclusion

From the analysis using ANOVA and Chi-square testing, we reach the following conclusions:

- **Educational Consistency**: The impact of online usage on daily routines and time management is consistent across gender lines, suggesting that digital habits affect the educational schedule of all students similarly.
- **Net Banking Dependency**: There is a significant association between gender and the impact of online usage on net banking.
- **Healthcare and E-commerce Independence**: There is no significant difference in the mean impact of online usage on healthcare or e-commerce based on gender.
- **Social and Mental Well-being**: The relationship between gender and the online usage impact on both mental health and lack of social commitment is independent.
- **Daily Routine and Time Management**: Gender does not significantly affect the mean impact of online usage on a student's daily routine or their time consumption.
- **Cyber Awareness**: Awareness of cyber fraud is independent of a student's mode of study suggesting that learning environments do not dictate levels of digital risk awareness.

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APPENDIX

A STUDY ON THE DOUBLE EDGED WORLD OF ONLINE AFFAIRS AMONG STUDENTS OF THIRUVANANTHAPURAM

QUESTIONNAIRE

Demographic Information

Name

Age

- Below 18
- 18-20
- 20-25
- Above 25

Area of residence

- Urban
- Rural

Gender

- Male
- Female

Mode of study

- Online
- Offline
- Hybrid

Education and Time consuming

Do you think online resources positively contribute to your academic performance?

- Strongly agree
- Agree
- Neutral

- Disagree

Do online learning makes it easier to manage study time compared to traditional methods?

- Very much easier
- Easier
- Neutral
- Not change

How many hours do you spend online per day?

- Less than 1 hr
- 1-2 hrs
- 3-4 hrs
- More than 5 hrs

Online activities that consume most of your time unnecessarily?

- Social media
- Online classes
- Online shopping
- Online streaming(movies/series)
- Others ----------

How often does online usage interfere with your daily routine?

- Always
- Sometimes
- Rarely
- Never

Health care and Scams

Have you ever used telemedicine(online health care services)?

- Always
- Sometimes
- Never

What is the main reason for using online health care services?

- Saves time
- Easy access to doctor
- Emergency use
- Lower cost

What type of online health care do you use most?

- Online doctor consultation
- Health app for fitness
- Online lab test booking

- Mental health support

Do scam calls and messages reduce trust in online services?

- Always
- Sometimes
- Never

Have you ever been a victim of of an online scam?

- Never
- One time
- Many time

Net banking and Lack of social commitments

How available online banking for you?

- Available anytime
- Available most of the time
- Rarely available
- Not available

How secure do you feel while using online banking?

- Very secure
- Secure
- Slightly secure
- Not sure

Do you agree online activities reduce student's participation in social activities?

- Strongly agree
- Agree
- Disagree
- Strongly disagree

Do you prefer online communication over real-life social interaction?

- Strongly prefer online
- Prefer online
- Prefer real-life interaction
- Strongly prefer real-life interaction

Do you agree online usage negatively affects students social commitments?

- Strongly agree
- Agree
- Neutral
- Disagree

Transportation and Mental health

How often do you use online transportation?

- Daily
- Weekly
- Occasionally
- Rarely

What are the reason for using online transportation?

- Convenience
- Time saving
- Over crowding in bus
- Easy cancellation and rescheduling

Are you aware of online transportation services India?

- Very much
- Somewhat
- No

Do you often compare yourself with others on social media ,leading to stress or low-self esteem?

- Yes frequently
- Sometimes
- Rarely
- Never

Have you ever experienced anxiety or depression due to online interaction like cyber bullying or negative comments?

- Many times
- One time
- Never

Ecommerce and Laziness

Are you aware of e-commerce platforms?

- Very much aware
- Slightly aware
- Not aware

How frequently do you use e-commerce platforms for productive purposes such as selling products, promoting services?

- Never
- Occasionally
- Frequently

- Very frequently

Do you think e-commerce saves time compared to offline shopping?

- Yes
- No
- Sometimes

Do you prefer online services instead of doing tasks physically?

- Always
- Often
- Sometimes
- Never

Do you ever feel online entertainment makes you less productive?

- Many times
- Some times
- Never

To what extent do online activities help or affect the following aspects of your daily life?

(1:Not at all – 5:To a great extent)

	1	2	3	4	5
Academic performance					
Health care					
Net banking					
Transportation					
E-commerce					
Mental health					
Lack of social commitment					
Time consuming					
Daily routine					